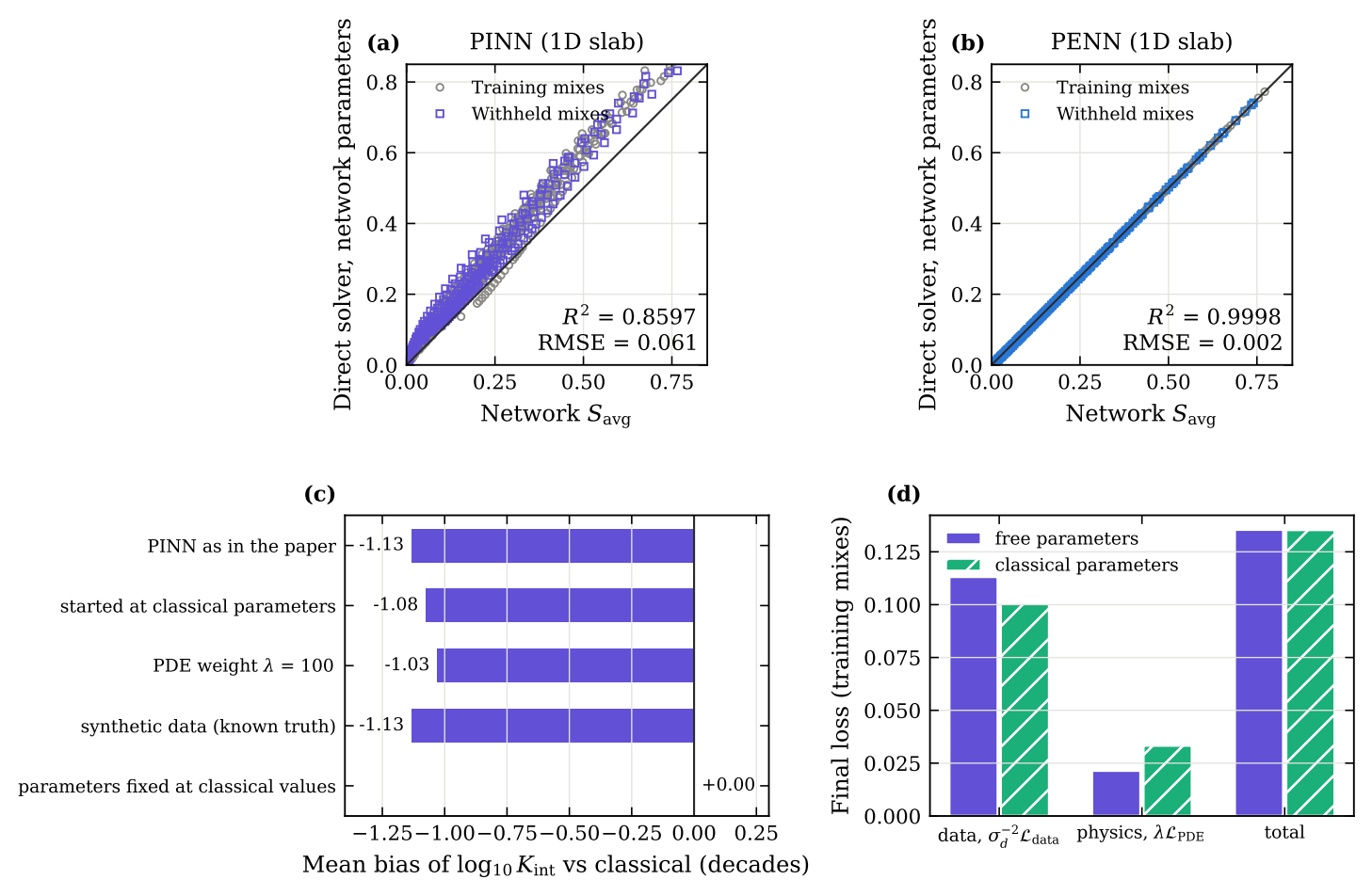

(a)
PINN (1D slab)
Training mixes
Withheld mixes
R^2 = 0.8597
RMSE = 0.061
Direct solver, network parameters
Network S_{avg}
(b)
PENN (1D slab)
Training mixes
Withheld mixes
R^2 = 0.9998
RMSE = 0.002
Direct solver, network parameters
Network S_{avg}
(c)
PINN as in the paper
started at classical parameters
PDE weight λ = 100
synthetic data (known truth)
parameters fixed at classical values
-1.13
-1.08
-1.03
-1.13
+0.00
Mean bias of $\log_{10} K_{int}$ vs classical (decades)
(d)
free parameters
classical parameters
Final loss (training mixes)
data, $\sigma_d^{-2}\mathcal{L}_{data}$
physics, $\lambda\mathcal{L}_{PDE}$
total